Evaluation and Clinical Validation in EHR

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Outline

- Why evaluation in healthcare differs from standard ML
- Clinical meaning of sensitivity, specificity, PPV, NPV
- Effects of class imbalance.
- ROC vs PR curves.
- How decision thresholds affect patient outcomes

Objective:
Learning how to properly evaluate DS models in healthcare

Why Evaluation in Medicine is Different

Errors have **different consequences** in healthcare:

Task	False Negative (FN)	False Positive (FP)
Sepsis early warning	Untreated sepsis	Extra monitoring/alert fatigue
ICU mortality prediction	Missed risk	Overtreatment
Cancer detection	Cancer remains untreated	Wrong treatment, invasive procedures

Sepsis

- A **life-threatening organ dysfunction** caused by an abnormal response to infection.
- Potentially leading to organ failure, septic shock or death (if not treated immediately).
- Common in ICU and emergency settings.
- High mortality (≈ 20–40% in severe cases).
- Time-critical: early treatment significantly improves survival.

ICU Mortality Prediction

- Identify patients at risk of death in the near future.
- Example: predict mortality 24–48 hours before onset.
- Enable preventive interventions.

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Which error is worse in cancer screening?

Cancer Detection

- Cancer **detection (diagnosis)** occurs when a patient **already has symptoms or suspicious findings**.
- Examples:
  - Detecting lung tumors in a CT scan
  - Classifying a suspicious lesion in pathology
  - Confirming malignancy after abnormal screening

Key characteristics	**Detection**
Population	Symptomatic or high-risk
Disease Prevalence	High
Goal	Confirm disease

Cancer Screening

- Cancer **screening** is testing **asymptomatic people** in the general population to detect disease early.
- Examples:
  - Breast Cancer screening with **mammography**
  - Colorectal Cancer screening with colonoscopy or FIT
  - Cervical Cancer screening with Pap smear

Key characteristics	**Screening**
Population	General population (mostly healthy)
Symptoms	None
Disease prevalence	Low
Goal	Early detection (detect disease before symptoms)

Why Evaluation in Medicine is Different

Errors have **different consequences** in healthcare:

Task	False Negative (FN)	False Positive (FP)
Cancer detection	Cancer remains untreated	Wrong treatment, invasive procedures
Cancer screening	Missed early cancer → later diagnosis	Unnecessary biopsies, anxiety, extra imaging

Which error is worse in cancer screening?

What metric matters most to evaluate a model?

Sensitivity and Specificity

- **Sensitivity** = TP / (TP + FN) = TP / Positives
- Ability to detect sick patients

  ↓ FN → ↑ Sensitivity

- **Specificity** = TN / (TN + FP) = TN / Negatives
- Ability to avoid false alarms

- Clinical models often require balancing these two metrics.

Key components:
- True Positive (TP)
- False Positive (FP)
- True Negative (TN)
- False Negative (FN)

Question

- If a test has 90% sensitivity and 90% specificity, what is the probability that a positive test means cancer?

→ Use the notebook “DSxH-2526-evaluation-metrics.ipynb” to answer this question.

Metric Priorities

- Screening usually prioritizes: **High sensitivity**
- Reason: Missing an early cancer is considered worse than doing extra follow-up tests.
- But specificity still matters because low prevalence can create **many false positives**.

Predictive Values

- Positive Predictive Value (PPV) = TP / (TP + FP)
- Negative Predictive Value (NPV) = TN / (TN + FN)

- Important: PPV strongly depends on disease prevalence.
- Low prevalence diseases often have low PPV.

Effects of class imbalance.

Effects of class imbalance

- Observed in the prediction of LOS problem:
  - Positive (long LOS) samples = 41.
  - Negative (short LOS) samples = 99.

- The Clinical Reality:
  - FN: patients predicted as short stay, but who actually stayed a long time.
  - In a ICU, FN are the "surprises" that cause logistical chaos.
  - "Long Stay" patient **consume more resources, require complex discharge planning, and are often higher risk.**

Effects of class imbalance

- Measuring accuracy is not enough!
- Compute other measures and interpret them.

ROC vs PR curves

ROC vs PR curves

- ROC (Receiver Operating Characteristic) Curve plots **True Positive Rate** (Sensitivity or Recall) against **False Positive Rate** (False Alarm Rate or 1 - Specificity).
- PR (Precision-Recall) Curve plots **Precision** (Positive Predictive Value) against **Recall** (Sensitivity).
- PR Curve ignores True Negatives and focuses entirely on how well you find the positives.

- Why is ROC not enough in imbalanced datasets?

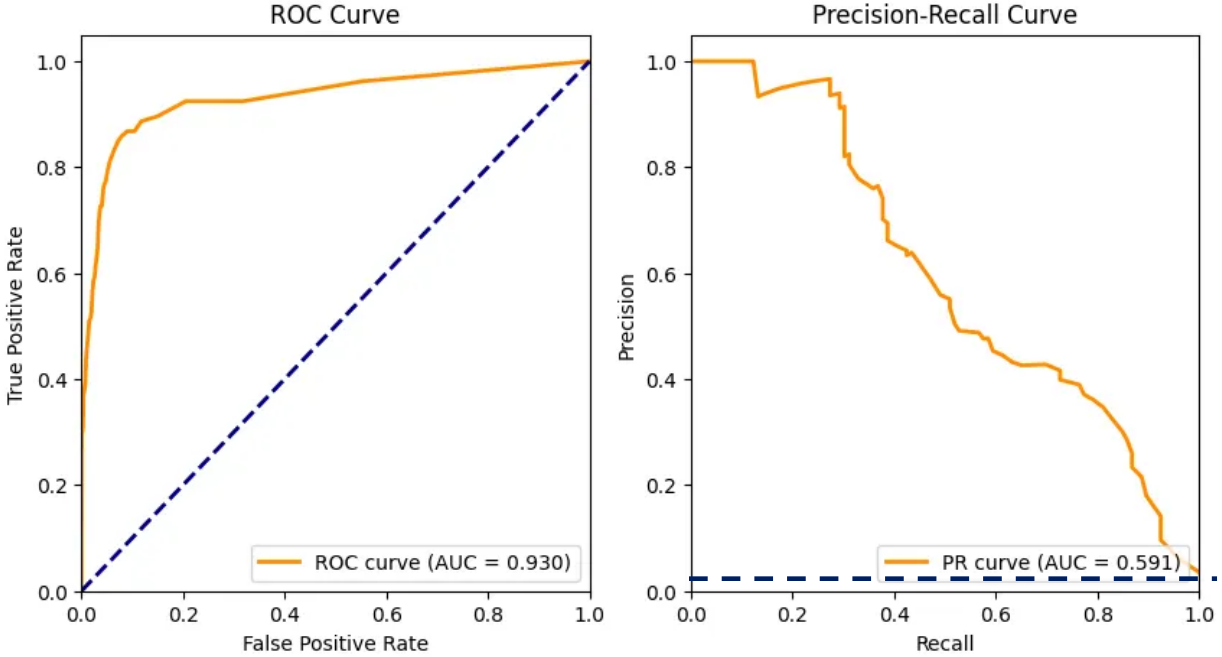
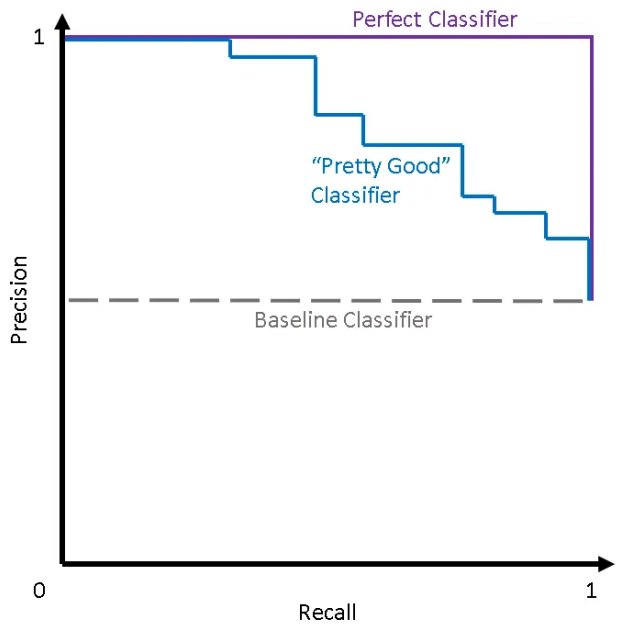


Figure: Evaluation of a classifier in an imbalanced dataset (10,000 samples and only 3% of samples belonging to the positive class).

Precision-Recall curve



This means the model found every single positive case without making a single mistake.

The Trade-off: As you try to increase your **Recall** (catching more cases), your **Precision** usually drops.

The Baseline (AUCPR of a random classifier): horizontal line representing the fraction of positive samples in the dataset.

- **AUC-PR (Area Under the Curve):** Value in [0-1].
  - Higher area means the model is better at maintaining high precision while increasing recall.

- **F1-Score:**
  - The harmonic mean of Precision and Recall: $F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
  - The point on the curve where you find the best balance between the two.

When to Use PR vs. ROC

Use ROC Curve (TPR vs FPR)	Use PR Curve (Precision vs Recall)
When classes are roughly **balanced**.	When classes are **heavily imbalanced** (e.g. rare disease).
When you care equally about False Positives and False Negatives.	When the **Positive class** is much more important than the Negative class.
When you want to see how well the model separates the two classes.	When you want to see how "pure" your positive predictions are.

Discussion Questions

- Is AUROC better than accuracy to evaluate a model?
- Is AUROC enough to evaluate medical models?

Decision Thresholds

Decision Thresholds

- Is it useful to change the threshold in the LOS prediction models?

Clinical Utility and Thresholds

- Predicted probabilities must be converted to decisions.

- Example: Sepsis alert threshold
- Low threshold → many alerts
- High threshold → missed cases

- Choosing thresholds requires **clinical judgment about consequences and workflow**.

Key Takeaways

- Evaluation metrics have clinical meaning
- Prevalence affects predictive values
- Clinical usefulness goes beyond accuracy

Data Science for Health